

EXPERIMENT 1

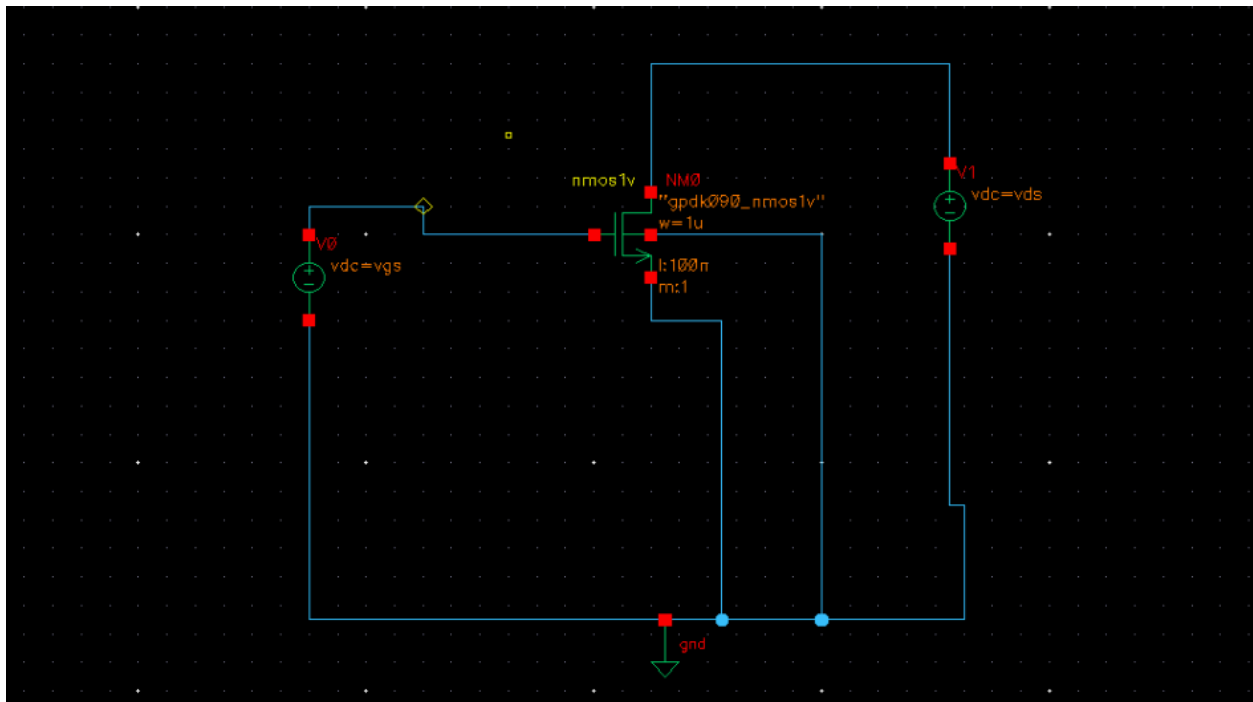
NMOS CHARACTERIZATION

Name: Bebin Sam

Roll Number: 2025BEC0007

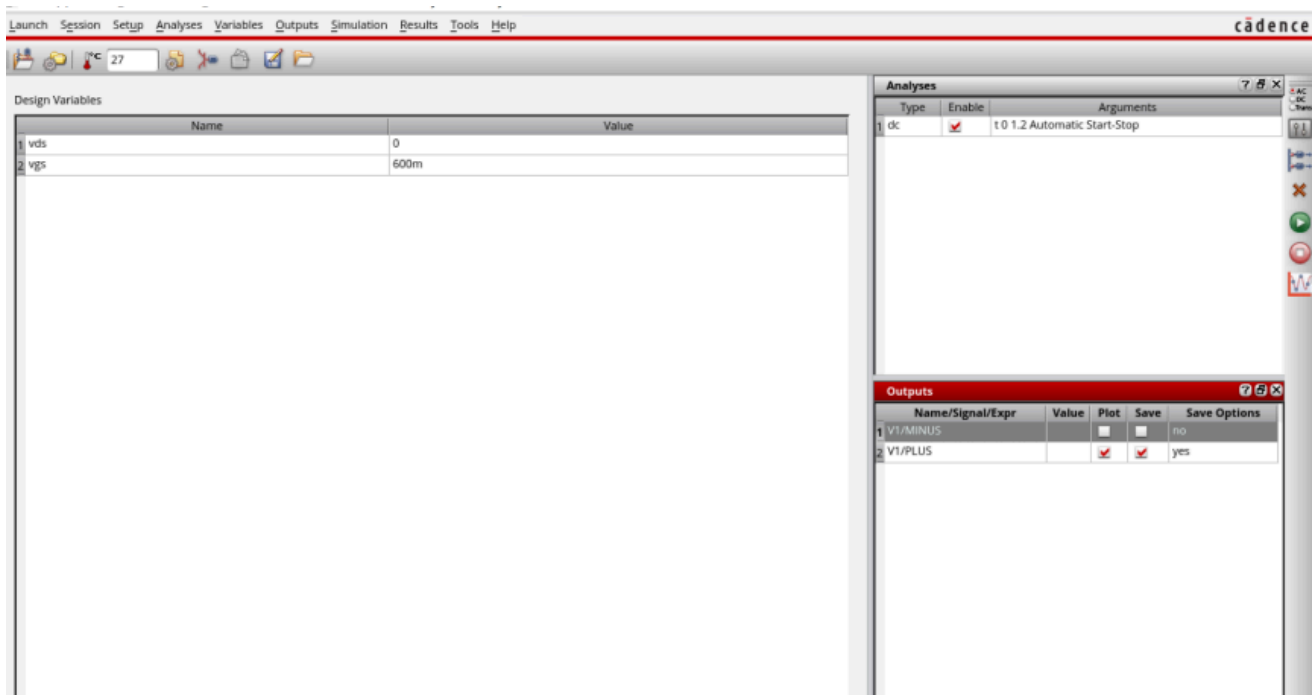
Github: https://github.com/Bebin-Sam/AnalogElectronicsLab_2025BEC0007

SCHEMATIC

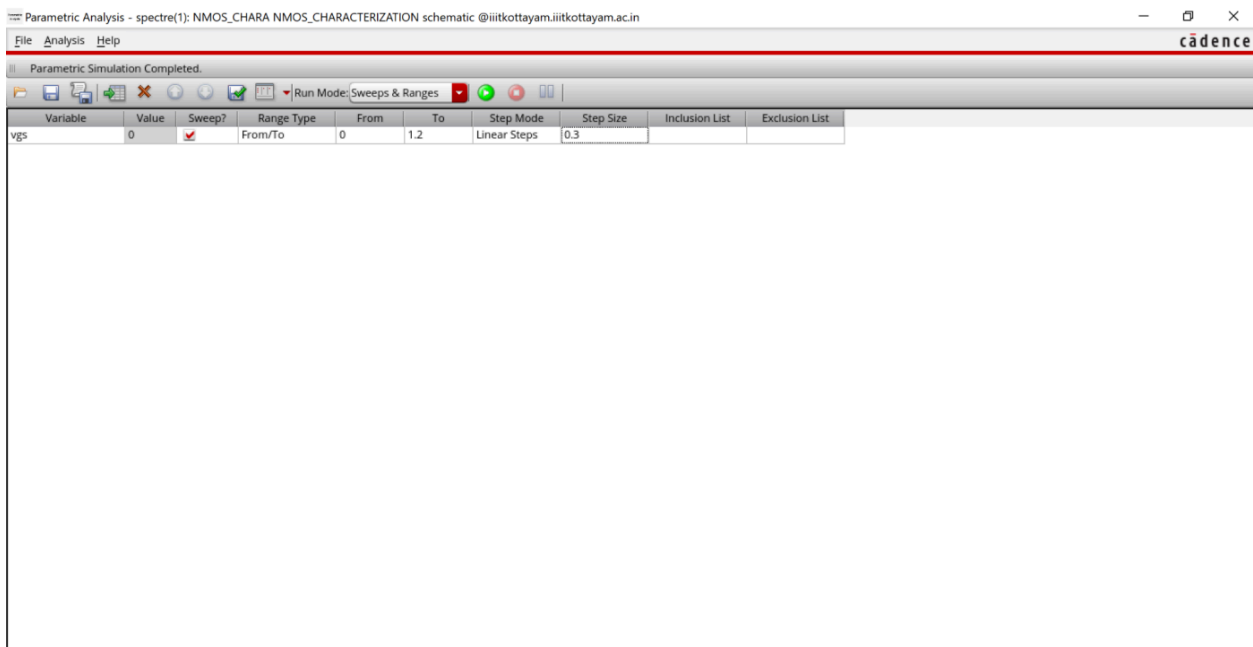


SIMULATION

1)DC Analysis Setup



2)Vgs Parametric Analysis

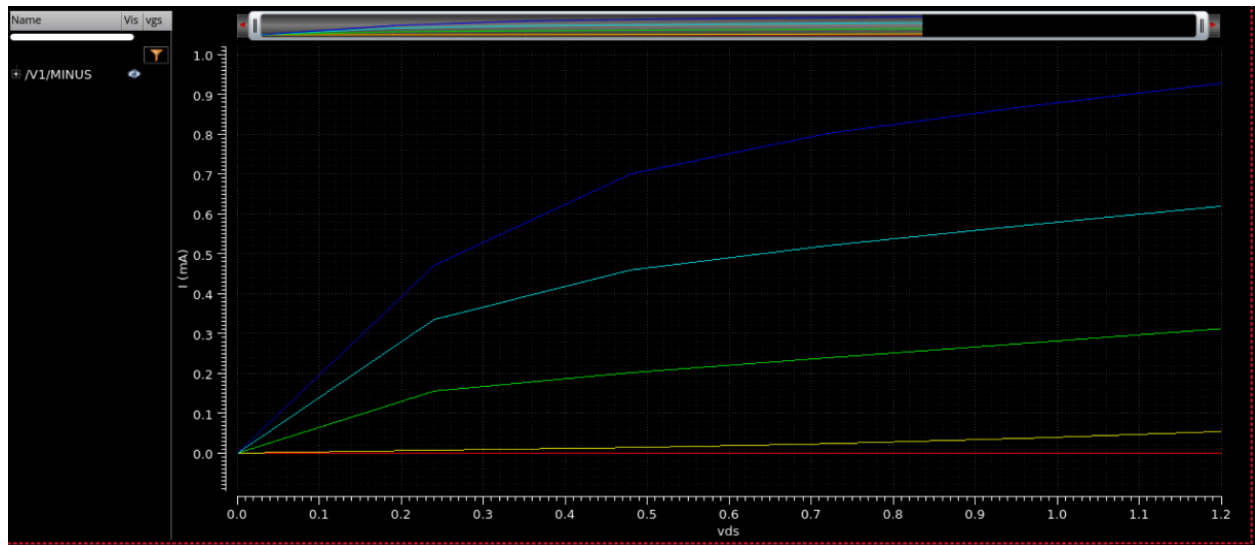


DC OPERATING POINT

signal	OP("/NM0" "?")
opdef	2.14748G
pwr	180.591n
qb	-225.151a
qbd	-350.222a
qbi	-225.151a
qbs	0
qd	565.421a
qdi	128.108a
qg	-332.143a
qgi	105.17a
qinv	48.6599u
qsi	-8.12739a
qsrco	-8.12739a
region	3
reversed	0
ron	7.97463M
rout	2.08771M
sdint	2.14748G
sdop	-1
self_gain	7.79505
tau1	NaN
tk	NaN
trise	NaN
ueff	20.7294m
vbs	0
vdb	1.2
vds	1.2
vdsat	42.1929m
vdsat_marg	NaN
vdss	42.1929m
vearly	314.19m
vfbeff	-908.859m
vgb	0
vnd	1.2

VI CHARACTERISTICS

1) I_D vs V_{ds}



2) I_D vs V_{gs}

